每日医学科研精读日报 | 2026年10月31日

一、今日前沿论文精选

1. A User-Centric Approach to the Development of a Mobile Yoga App for Patients With Thoracic Cancer and Their Family Caregivers.

- 发表期刊 / 影响因子:JMIR formative research
- 发表时间:2026-Oct-30
- 第一/通讯作者:Kathrin Milbury, Olivia Chang, Meagan Whisenant, Jason R Bentley, Scherezade K Mama
- 核心结论:We recently completed the efficacy testing of an instructor-led yoga program for patients with thoracic cancer undergoing radiotherapy and their family caregivers. Translating an instructor-led intervention to a mobile app-based delivery model may facilitate the implementation of this yoga program i…
- 原文DOI / 链接:10.2196/89649

2. Decoding Lung Cancer with Artificial Intelligence: A Path to Personalized Diagnostics and Therapy.

- 发表期刊 / 影响因子:Journal of environmental pathology, toxicology and oncology : official organ of the International Society for Environmental Toxicology and Cancer
- 发表时间:2026
- 第一/通讯作者:Tafadzwa Carl Masuka, Cecileya Jasmin Meshak Dhanashekaran, Vinod Prabhu Venugopal
- 核心结论:Non-small-cell lung cancer (NSCLC) represents the majority of lung cancer cases and remains the leading cause of cancer-related mortality worldwide. Early detection, accurate staging, and individualized therapy are crucial for improving patient survival. Recent advances in artificial intelligence (A…
- 原文DOI / 链接:10.1615/JEnvironPatholToxicolOncol.2026061968

3. Comprehensive Characterization of Crotonylation-Related Long Non-Coding RNAs Reveals a Prognostic Signature and Functional Landscape in Bladder Cancer.

- 发表期刊 / 影响因子:Journal of environmental pathology, toxicology and oncology : official organ of the International Society for Environmental Toxicology and Cancer
- 发表时间:2026
- 第一/通讯作者:Wuguang Zhang, Kaifeng Zheng, Min Peng, Jing Zhang, Xien Wang
- 核心结论:Lysine crotonylation (Kcr), a novel post-translational modification, has emerged as a key epigenetic regulator implicated in diverse cellular processes and tumorigenesis. However, the biological and clinical relevance of crotonylation-related long non-coding RNAs (lncRNAs) in bladder cancer remains…
- 原文DOI / 链接:10.1615/JEnvironPatholToxicolOncol.2025060257

4. The role of microbial enzymes in intervention strategies for metabolic diseases targeting the gut microbiota.

- 发表期刊 / 影响因子:Science China. Life sciences
- 发表时间:2026-Oct-27
- 第一/通讯作者:Yuhua Gao, Xiaoxiang Gao, Kai Wang, Jie Qiao, Changtao Jiang
- 核心结论:Metabolic diseases-including obesity, metabolic dysfunction-associated steatotic liver disease (MASLD), type 2 diabetes mellitus (T2DM), atherosclerosis (AS), and polycystic ovary syndrome (PCOS)-pose a significant global health crisis due to their rapidly increasing prevalence. The therapeutic pote…
- 原文DOI / 链接:10.1007/s11427-026-3413-3

5. Association between the Inflammatory Atherosclerosis Index and Non-Alcoholic Fatty Liver Disease: NHANES 2017-2020.

- 发表期刊 / 影响因子:Arquivos brasileiros de cardiologia
- 发表时间:2026
- 第一/通讯作者:Yun Jiang, Jingying Xie, Ying Dong
- 核心结论:The Non-alcoholic fatty liver disease (NAFLD) prevalence is increasing globally and is closely associated with dysregulated lipid metabolism and inflammation. However, the relationship between NAFLD and the inflammatory atherosclerosis index (IAI), which reflects these processes, is unclear. To inve…
- 原文DOI / 链接:10.36660/abc.20260068

6. Rethinking Neural Thickets Without Weight Perturbations

- 发表期刊 / 影响因子:arXiv Preprint
- 发表时间:2026-10-29
- 第一/通讯作者:Nicolas Dufour, Elouan Gardès, Amil Dravid, Patrick Pérez, Loic Landrieu
- 核心结论:The Neural Thickets hypothesis posits that pretrained language models lie in dense neighborhoods of task-specific experts in weight space, which random Gaussian perturbations can uncover. We reproduce the reported gains, but show that they do not require this geometric explanation. Perturbing a sing…
- 原文DOI / 链接:https://arxiv.org/abs/2610.40649v1

7. Read, Allocate, Route: Role-Coupled Tokenization for Pixel-Space Generative Transformers

- 发表期刊 / 影响因子:arXiv Preprint
- 发表时间:2026-10-29
- 第一/通讯作者:Jiongfan Yang, Weichen Fan, Haiwen Diao, Zhongang Cai, Ziwei Liu
- 核心结论:Pixel-space generative Transformers typically use one token per image patch, so capturing finer detail requires more tokens and makes global attention increasingly expensive. We observe that finer detail does not necessarily require more global communication: low-frequency structure benefits from gl…
- 原文DOI / 链接:https://arxiv.org/abs/2610.40647v1

8. Propose-Filter-Release: Scaling Noise to Typical Sensitivity via Private Filtering

- 发表期刊 / 影响因子:arXiv Preprint

- 发表时间:2026-10-29
- 第一/通讯作者:Moshe Shenfeld, Eliad Tsfadia, Katrina Ligett
- 核心结论:An estimator can be accurate and stable on typical data, yet sensitive to a few possible records. Differentially private mechanisms commonly scale their noise to the worst-case influence of a single record, which lets these records dictate the noise even when the remaining data would allow far less⋯.
- 原文DOI / 链接:https://arxiv.org/abs/2610.40654v1

二、科研公众号干货

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三、每日高分经典精读

Stereotactic Body Radiation Therapy for Hepatocellular Carcinoma: Current Trends and Future Directions

- 发表期刊 / 影响因子 / 总引用量:International Journal of Radiation Oncology, Biology, Physics / IF=7.0 / 引用量:200+
- 研究背景与里程碑意义:SBRT has emerged as a definitive local treatment for HCC, but optimal dosing, patient selection, and integration with systemic therapies remain active areas of investigation.
- 核心实验设计与方法:Comprehensive review of clinical trials and retrospective studies of SBRT for HCC, including dose-escalation studies, toxicity analyses, and combination strategies.
- 关键结论与行业影响:SBRT achieves local control rates of 80-95% for early-stage HCC with low toxicity. Challenges include BCLC stage-specific outcomes, liver function preservation, and optimal sequencing with immune checkpoint inhibitors.
- 创新点:Systematic framework for SBRT dose selection based on tumor size, location, and liver function; proposed risk-adapted fractionation schedules.
- 原文DOI / 链接:https://doi.org/10.1016/j.ijrobp.2023.01.045

四、当日总结

今日共精选8篇最新研究+1篇经典精读。关注方向:A User-Centric Approach to the Development of a Mobile Yoga App for Patients With Thoracic Cancer and Their Family Caregivers.
